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**Offers you Enterprise Application Development Package,**
**which includes:**

- Core Java
- SQL
- Web Technology
- J2EE
- Basic of Frameworks (Spring & Hibernate)
- React JS

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# Course Contents are as follows:

## Core Java Course Content

- **Introduction to Programming**
  1. Introduction to java
  2. JDK installation
  3. Keywords, Identifiers, variables
  4. Operators
  5. Method/Functions
  6. Flow Control Statements
  7. Arrays
  8. Strings
  9. Interactive programs in java using Scanner
- **Object Oriented Programming System**
  1. Classes and Objects
  2. Object creation
  3. Reference variable
  4. Global and local variables
  5. Constructors
  6. Aggregation
  7. Composition
  8. Inheritance ***
  9. Method Overloading
  10. Method Overriding
  11. Abstract classes
  12. Interfaces
  13. Typecasting
  14. JVM architecture

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15. Polymorphism
16. Abstraction
17. Java packages
18. Access Specifies

- **Java Built-in Packages**

1. Classes and Object
2. String, String Buffer and String Builder
3. Exception Handling
4. Threads and multithreading
5. Wrapper Classes
6. Data Structures
7. JAVA COLLECTION FRAMEWORKS***
8. File Handling
9. Serialization
10. Garbage Collector
11. Encapsulation

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## SQL Course Content

- **Introduction**

1. What Is Database?
2. What is Database Management System (DBMS)?
3. What is Relational Model?
4. Introduction to RDBMS
5. Brief on E.F CODD .Native Application

- **Datatypes and Constraints**

1. What are Datatypes?

2. Types and Examples.
3. How to use?
4. What are Constraints?
5. Types and Examples.
6. How to use?

- **Statements in SQL**
  1. Data Definition Language (DDL)
  2. Data Manipulation Language (DML)
  3. Transaction Control Language (TCL)
  4. Data Control Language (DCL)
  5. Data Query Language (DQL)

- **Software installation**
  1. Installing and set up of software
  2. Working on Oracle 10g.
- **Data Query Language (DQL)**
  1. Select
  2. From
  3. Where
  4. Group By
  5. Having
  6. Order By

- **Operators**
  1. Types and Examples
- **Functions in SQL**
  1. Single Row Functions
  2. Multi Row Functions

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- Max ()
- Min ()
- Sum ()
- Avg ()
- Count ()

- **Sub Query**
  1. Introduction to Sub Query
  2. Working of Sub Query
  3. Query Writing and Execution
  4. Types of Sub Query
     - Single Row Sub Query
     - Multi Row Sub Query
     - Nested Sub Query.
- **Pseudo Columns**
  1. Introduction on Pseudo Columns
  2. ROWID
  3. ROWNUM
  4. Working and Usage.
- **JOINS**
  1. What Is Join?
  2. Types of Joins.
  3. Cartesian Join
  4. Inner Join
  5. Outer Join
  6. Self-Join
  7. Queries and Examples.

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- **Co- Related Sub Query**
    1. Working and Examples

- **Data Definition Language (DDL)**
    1. Create
    2. Rename
    3. Alter
    4. Truncate
    5. Drop

- **Data Manipulation Language (DML)**
    1. Insert
    2. Update
    3. Delete

- **Transaction Control Language (TCL)**
    1. Commit
    2. Save point
    3. Rollback

- **Data Control Language (DCL)**
    1. Grant
    2. Revoke
- **Normalization**
    1. Introduction to Normalization
    2. Types of Normal Forms
    3. Examples.
- **E R Diagrams**

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1. Introduction to ERD
2. Examples.

# Web Technologies Course Content

## HTML:

- **Introduction to HTML5**
  1. How Webworks, what is the Internet, What is Server, Browser, etc,
  2. What is Website, Web Pages, Web Application, Web technology, Web Browser, Web Server
  3. Static and Dynamic pages
  4. SPA & MPA
  5. History of HTML
- **HTML5 Basics**
  1. Understand the structure of an HTML page
  2. What are HTML Tags, Types of Tags
  3. HTML Elements, Grouping Elements
  4. HTML Attributes, Core Attributes
  5. HTML Basic Tags
  6. HTML Formatting Tags
- **List**
  1. Unordered List
  2. Ordered List
  3. Nested List
  4. Definition List
- **Tables**
  1. Understand the structure of an HTML table
  2. Table with thead, tbody, tfoot and caption
  3. Applying Table Border

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4. Applying Background and Foreground fills
5. Changing Cell Padding, Spacing and Alignment

- **Anchors and Hyperlinks**
  1. Link to another site
  2. Link to an anchor
  3. Link to a page on the same site
  4. Link that dials a number
  5. Open link in new tab/window
  6. Link that runs email
  7. Add hyperlinks to list items and table contents
- **Image Handling**
  1. Understand the role of images in web pages
  2. Learn to add images to web pages
  3. Learn to use images as hyperlinks
- **HTML Forms**
  1. Understand the role of forms in web pages
  2. Form Elements
  3. Input Types
  4. Input Attributes
  5. New Form Elements of HTML5
  6. Form Validation
- **Sectioning Element**
  1. Semantics
- **HTML Media**
  1. Audio
  2. Video
- **HTML Graphics**
  1. Canvas

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2. SVG

- **HTML5 – Client-Side Storage**

  1. **Introduction to HTML5 Client-Side Storage**
  2. **Types of Client-Side Storage**

## CSS:

- **Introduction to CSS3**

  1. **What Is CSS3?**
  2. **CSS Syntax**
  3. **The History of CSS**
  4. **How to Add CSS:**
     - **Inline Styles**
     - **Internal Styles**
     - **External Files**

- **CSS Selectors**

  1. **Simple selectors:**
     - **Universal Selector (*)**
     - **Element Selector**
     - **Id Selector (#)**
     - **Class Selector(.)**
  2. **Combinator selectors:**
     - **Descendant Selector**
     - **Child Selector**
     - **Adjacent Sibling Selector**
     - **General Sibling Selector**
  3. **Attribute Selectors**
  4. **Pseudo-class selectors**

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5. Pseudo-elements selectors

- **Working with Text and Fonts**
  1. Text Formatting
  2. Text Effects
  3. Text Shadow
  4. Text Outline
  5. Text Stroke
  6. Fonts
  7. The @font-face Rule
  8. Using Web Fonts: Google, Typekit, etc
  9. Font Properties

- **CSS Background, Colors and Gradients**
  1. CSS Background
  2. Colors, RGB and RGBA
  3. Opacity Property
  4. HSL and HSLA
  5. Linear Gradients
  6. Radial Gradients
- **Border, Display and Box Effects**
  1. Borders
  2. Rounded Corners
  3. Applying Shadows in border
  4. Image Borders
  5. Resizing
  6. Visibility
  7. Outline Offsets
  8. Box Shadow

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9. Display Property

- **Box Model, Z-index and CSS Rule**

1. CSS Box Model: Text, Padding, Border, Margin
2. CSS !important
3. z-index

- **Position, Overflow, float and Border image**

1. Position Property
2. Overflow Property
3. Float Property
4. Border Image

- **Layout: Columns, Flexible Box**

1. CSS Website layout (Grid/Columns)
2. flex box
3. flex container
4. flex items
5. flex responsive

- **Transforms**

1. 2D Transforms:
   - translate(), rotate(),
   - scale(), screw(), matrix()
2. 3D Transforms :
   - rotateX(),
   - rotateY()

- **Transitions**

1. Transition
2. transition-delay

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3. transition-duration
4. transition-property
5. transition-timing-function

- **Animations**
  1. @keyframes
  2. animation-name
  3. animation-duration
  4. animation-delay
  5. animation-iteration-count
  6. animation-direction
  7. animation-timing-function
  8. animation-fill-mode
  9. animation

- **CSS Media Queries**
  1. Targeting Device Capabilities: Width, Screen Size, Color Depth, etc.
  2. Responsive Images and the viewport Metatag
- **Vendor Prefixes**
  1. Current status of -moz, -o, etc.
  2. When and Why to Use

## JavaScript :

- **Introduction to JS**
  1. What is JavaScript
  2. History of JavaScript
  3. Advantages & Disadvantages of JavaScript
  4. Environment Setup

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5. How does Webpages Work
6. JavaScript Engines
7. JavaScript Workflow
8. Basic Introduction to JavaScript Call Stack

- **JavaScript Fundamentals**
  1. How to include JavaScript in an HTML document
  2. The JavaScript syntax
  3. Hello World Program
  4. Variables (var, let, const)
  5. Reserved Words
  6. Data Types:
     - Primitive (number, string, Boolean, null, undefined, BigInt, Symbol)
     - Object Reference (Array, Object, Function, Math, Date)
- **JavaScript Operators:**
  1. Arithmetic
  2. Assignment
  3. Comparison
  4. Conditional
  5. Statements
  6. Control Structures
  7. Loops
- **JavaScript Functions**
  1. What are Functions
  2. Function syntax
  3. function(parameters)
  4. return
  5. Function Declarations vs. Expressions

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6. Defining Functions (AN, NF,FE,IIFE)
7. Functions with default parameters
8. Function with Arguments Object
9. this object
10. Scope
11. Functions Calling Other Functions
12. Call back Function
13. Reviewing Functions
14. Closures, Recursion
15. Hoisting (Both Variable & function)
16. Global & Local Variables
17. Higher Order Function,
18. Pure Function/First class function
19. Constructor Function

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- **JavaScript Arrays**

1. Introduction to Arrays
2. Creating the Array with literals and constructor
3. Basic Array Operations
4. Iterating and transforming Arrays
5. Array Prototype Methods:

   push(), pop(), shift(), unshift(),indexOf(),includes(),Splice(),Slice(),forEach(),map(),filter(), Sort(), reduce(), lastIndexOf(), reduceRight(), find(), findIndex(), flat(), flatMap(),Some(), every(),reverse()

- **Array Functions:**

  - Array.of()
  - Array.from()
  - Array.isArray()
  - Complex Array

- ## JavaScript Objects

    1. **Introduction to Objects**
    2. **Creating Objects with literals and constructor**
    3. **Nested Objects**
    4. **Dot v/s Bracket Notation**
    5. **Object Methods: Object.assign(),Object.create(),Object.defineProperty(),Object.defineProperties() , Object.entries(), Object.freeze()Object.keys(), Object.is(), Object.seal(), Object.values()**
    6. **Working with Object Instances**

- ## Iterations & Looping

    1. **What are iterators**
    2. **for loop, for...of, for…in**
    3. **Looping Arrays, Breaking and Continuing**
    4. **Looping Backwards and Loops in Loops**

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- ## Date, Math Objects and JSON

    1. **Date Object**
    2. **Time delays**
    3. **Math Object**

- ## JavaScript Runtime behaviour

    1. **The JavaScript Engine and Runtime**
    2. **Execution Contexts.**
    3. **The Call Stack**
    4. **Scope and The Scope Chain**

- ## String Methods, this keyword, strict mode and types of errors

  1. **String Methods: toLowerCase(), toUpperCase(), charAt( position ), indexOf( searchString, position ), concat(…strings), includes(search String, start position),replace(search Value, replaceValue), split( separator, limit), substr( start, length), search(value), trim()**
  2. **this keyword**
  3. **Strict mode: usestrict**
  4. **Types of Errors**

- ## Browser Object Model (BOM)

  1. **What is BOM**
  2. **window Object**
  3. **Screen**
  4. **Location**
  5. **History**
  6. **Navigator**
  7. **Popup Alert**
  8. **BOM properties and Objects**

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- ## Document Object Model (DOM)

  1. **What's the DOM**
  2. **DOM Manipulation**
  3. **DOM Methods**
  4. **DOM Properties**
  5. **Selecting, Creating, and Deleting Elements**
  6. **Styles, Attributes and Classes**

- ## DOM Event Handling & CSS Styling

  1. **What are Events**
  2. **Event Handling**

3. Types of Events
4. Event Handlers
5. Lifecycle of DOM Events
6. Event Propagation:
    - Bubbling and Capturing
    - Event Propagation
7. Event Delegation
8. Handling an "Esc" Keypress Event
9. Handling Click Events,
10. addEventListeners()
11. Manipulating CSS Styles
12. Efficient Script Loading: defer and async

- **Working with Form Object**

1. Form Object Properties, Methods & Event Handlers
2. Text-Related Objects
3. Button Objects
4. Check Box and Radio Objects
5. Select Objects
6. Validate Data and Form Submission

- **JavaScript Object Notation (JSON)**

1. JSON Introduction & Syntax
2. Why JSON is used
3. Data Types
4. JSON Parse and Stringify
5. JSON Objects and Arrays

- **JavaScript Debugging**

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1. Debugging the JavaScript code using:
    - console.log
    - debugger

- **Web API's**
    1. Geolocation
    2. Web Storage

## ECMAScript:

- **Introduction to ECMAScript**
    1. Introduction to ECMAScript
    2. Difference b/w JavaScript & ECMAScript
- **keywords & Symbol**
    1. let, const
    2. Difference b/w var, let & const
    3. Symbol
- **Arrow Functions & Template Strings**
    1. What are Arrow Functions
    2. Arrow function Syntax
    3. Arrow Functions (Implicit/Explicit) ways
    4. Regular Functions vs. Arrow Functions
    5. Template Strings
    6. Default function arguments

- **Destructuring**
    1. What is Destructuring?
    2. Array Destructuring
    3. Object Destructuring

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4. Destructuring with Complex Array, Nested Array, Swapping
5. Destructuring with Nested Objects

- **Spread & Rest operators**
  1. Spread Operator (...)
     - Spread Operator for Arrays
     - Spread Operator in function calls
     - Spread Operator for Objects
  2. Rest operators (...)
     - Rest parameters with functions
     - Rest Pattern and Parameters

- **Advanced ES6 Operators**
  1. Short Circuiting (&& and ||)

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- **JavaScript Data Structures**
  1. Map Operators
  2. Map Fundamentals & Iterations:
     - new Map()
     - set, get, has, delete
  3. Set Operators
  4. Set Fundamentals & Iterations:
     - new Set()
     - size, add, delete, clear, has
  5. Weak Maps and Weak Sets
  6. new WeakMap() / WeakSet()
     - add
     - remove

- ## Modules in JavaScript

    1. An Overview of Modules in JavaScript
    2. Exporting and Importing in ES6 Modules
    3. The Module Pattern
    4. CommonJS Modules
    5. A Brief Introduction to the Command Line
    6. Introduction to NPM
    7. Bundling With Parcel and NPM Scripts
    8. Configuring Babel and Polyfilling
    9. Exporting a Module
    10. Named Exports
    11. Default Exports
    12. Importing a Module
    13. Importing Named Exports
    14. Importing Default Exports

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- ## Object Oriented Programming in JavaScript

    1. OOP in JavaScript
    2. Constructor Functions and the new Operator
    3. Prototypes
    4. Prototypal Inheritance and The Prototype Chain
    5. Prototypal Inheritance on Built-In Objects
    6. ES6 Classes
    7. Inheritance
    8. Setters and Getters
    9. Static Methods
    10. Inheritance Between "Classes": Constructor Functions, ES6 Classes

- ## Asynchronous JavaScript, Ajax, Promises

1. synchronous JavaScript, AJAX and APIs
2. AJAX Call: XMLHttpRequest
3. How the Web Works
4. Requests and Responses
5. Promises and the Fetch API
6. Consuming Promises
7. Chaining Promises
8. Welcome to Callback Hell
9. Handling Rejected Promises
10. Throwing Errors Manually
11. Asynchronous Behind the Scenes:
12. The Event Loop
13. Consuming Promises with Async/Await
14. Returning Values from Async Functions

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## J2EE Course Content

- **JDBC**

1. Introduction to JDBC
2. JDBC Drivers
3. Database URL
4. JDBC as an API
5. Programs on JDBC
6. Static queries
7. Dynamic queries

- **JPA with Hibernate**

1. Introduction to Hibernate
    - Problems with JDBC & Need of Hibernate

- ORM Introduction and its need
- Hibernate with JPA
- The Persistence Life Cycle

2. Hibernate concepts
    - ID and Primary key
    - ID generation methods
    - Transactions
    - POJO and Data Layer
    - Crud Operations
3. Java Persistence Query Language
    - Select clause
    - From clause
    - Where clause
    - Aggregation functions
4. Hibernate Mappings
    - One-To-One
    - One-To-Many
    - Many-To-One
    - Many-To-Many
    - Mapping with List, Map and Set

- **Design Patterns**

1. Design Patterns vs Architectural patterns vs Architectural Style
2. Architectural Style
    - REST
3. What are Design Patterns?
4. Types of Design patterns
5. DAO
6. DTO
7. Creational Design Patterns

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8. Structural Design Patterns
9. Behavioral Design Patterns

- **Servlets**

1. Introduction
2. Web Server as an application
3. Installation of Tomcat Webserver
4. Creation of servlet
5. WAR file: Build and deployment process
6. Web URL structure
7. Query String
8. HTTP & HTTPS Protocols
9. Elements of HTTP Request & HTTP Response
10. HTTP Methods
11. Difference between the GET & POST method
12. Difference between other HTTP methods
13. Servlet Container
14. XML configuration and 'web.xml'

- **JSP**

1. Introduction
2. Difference between Servlets and JSP

## Spring Course Content

- **Spring Core**

1. IOC Container
2. Beans
3. Dependency Injection
4. Wiring beans
5. Bean Container

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6. Lifecycle of Bean
7. Scope of Bean
8. Annotations
9. XML, Annotation-based and mixed configurations

- **Spring MVC**

1. Introduction to Spring MVC / WEBMVC
2. Spring MVC as Monolithic Application
3. Flow of Spring MVC
4. Dispatcher Servlet
5. Controller Classes
6. Annotation based controller configuration
7. Redirect, Forward & Include
8. Attributes and Mappers
9. Cookies
    - Introduction
    - Persistent & Non-Persistent Cookies
10. Session
    - Introduction
    - Advantages of session
11. Handler Mappings
12. Resolving Views (View Resolver)

- **Spring REST**

1. Introduction
2. Need of Spring REST
3. Spring REST as Web Service
4. Service Oriented Architecture (SOA)
5. Why HTTP, XML and JSON?
6. Introduction to XML
    - Why XML?

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7. Introduction to JSON
    - JSON Syntax
    - JSON Data & Values
    - JSON Objects
    - JSON Arrays
    - JSON vs XML
    - JSON Marshalling and Un-Marshalling using Jackson
8. Spring REST Configuration using XML files
9. Spring REST Model Classes
10. Spring Restful web service End points
11. Spring Restful web service Controller class
12. Spring REST programs

- **Spring Boot**
    1. Introduction to Spring Boot
    2. Installation of Spring Tool Suite
    3. Starter Dependencies
    4. Deploy a Spring Boot WAR into Tomcat Server

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## React JS Course Content

- Basics of Javascript and ECMAScript
- Introduction to React
- Role of NodeJs and NPM in React
- Introduction and Implementation of Technologies in Front End Development
- Installation and Benefits of VS Code
- Installation of React plugins in IDE
- Installation of Node, and NPM

- Installation of React
- Data-flow in React
- How to create NPM package for React
- React Component : use and benefits
- How to render view in React
- Different Methods to Render view and Components
- Es6 with React
- JSX
    1. What is JSX
    2. Why JSX
    3. Rules of JSX implementation
    4. Components with and without JSX
- React Props and State
- React Prop in practice
- React State in practice
- React Developer Tool
- Event Handling
- More about React State: Counter App
- Handling Lists
- Handling User Input
- Styling Your Component
- Basics of React Interview Questions
- Introduction of Bootstrap
- Installation of Bootstrap
- How to implement theme in react
- How to create layout based structure in React

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- One Project based on React Js
- Lifecycle methods
- Hooks
    1. Introducing Hooks
    2. Using the State Hook
    3. Using the Effect Hook
    4. Rules of Hooks
    5. Building Your Own Hooks
    6. Hooks API Reference
- Fetching data
- Example app
- Diving deeper interview questions

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